



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination I
Academic Year: 2025-26 (Semester – I)

Class: TY B.Tech.	Subject: Database Management System	Subject Code: PCC-IT304/PCC-IT305
Date: 19.08.2025	Time: 15:00 to 15:45 pm	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Use of non-programmable calculator is permitted.

Q.N.		Marks	CO	BT
Q.1	Consider the following relational database schema consisting of the four relation schemas: passenger (pid, pname, pgender, pcity) agency (aid, aname, acity) flight (fid, fdate, time, src, dest) booking (pid, aid, fid, fdate) Give an expression in Relational Algebra for each of the following queries. a) Get the complete details of all flights to New Delhi. b) Get the details about all flights from Chennai to New Delhi. c) Find only the flight numbers for passenger with pid 123 for flights to Chennai before 06/11/2020. d) Find the passenger names for passengers who have bookings on at least one flight. e) Find the passenger names for those who do not have any bookings in any flights.	10 M	CO2	BT5
Q.2	a) Explain Instances and Schemas with suitable example?	05 M	CO1	BT2
	b) Describe Query Processing with suitable example?	05 M	CO3	BT2

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	demonstrate a comprehensive understanding of fundamental database concepts, including data models, schema architecture, and relational database principles.
CO 2	apply database design principles, such as entity-relationship modeling and normalization, to design efficient and scalable database schemas that meet specified requirements.
CO 3	acquire proficiency in writing complex SQL queries for data retrieval, manipulation, and administration tasks, as well as understanding the underlying principles of SQL optimization.
CO 4	demonstrate the ability to manage transactions effectively, including understanding concurrency control mechanisms, ensuring data consistency, and implementing recovery techniques.
CO 5	gain knowledge of database security principles and mechanisms, including access control, authentication, authorization, encryption, and auditing.
CO 6	develop skills in optimizing database performance through techniques such as indexing, query optimization, and database tuning.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
ISE -I (Semester-I) 2025-26

TY
Operating System

Information Technology
PCC - IT303

Date: 20/08/2025

Time: 15:00 – 15:45

Max. Marks: 20

Q. N	Question	CO, BT	Marks
Q. 1	Explain the main functions of an Operating System. Distinguish between a monolithic kernel and a microkernel with suitable examples.	CO1,2 BT 1	6

Q.2 Consider the following processes:

	Process Arrival Time	Burst Time
P1	0	5
P2	2	3
P3	4	8

- (i) Draw the Gantt chart using First Come First Serve (FCFS) scheduling.
(ii) Calculate the turnaround time and waiting time for each process.
(iii) Find the average turnaround time and average waiting time.

CO2
BT 2-5 7

Q.3 Consider the following set of processes with corresponding arrival times and burst times:

Processes	P1	P2	P3	P4	P5
Arrival time	0	3	5	7	10
Burst time	6	10	8	5	6

Draw the Gantt chart considering the Round Robin scheduling policy with time quantum = 4 units. Calculate individual turnaround time and average waiting time.

CO2
BT 2-5 7

ALL THE BEST

Bloom's Taxonomy

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

Course Outcomes:

- CO1 Understand the role of operating system as System software.
- CO2 Compare the various algorithms and comment about performance of various algorithms used for management of memory, CPU scheduling, File handling and I/O operations.
- CO3 Apply various concept related with Deadlock to solve problems related with Resources allocation, after checking system in Safe state or not.
- CO4 To appreciate role of Process synchronization towards increasing throughput of system.
- CO5 Describe the various Data Structures and algorithms used by Different OSs like Windows XP , Linux and Unix pertaining with Process , File , I/O management.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Information Technology
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Software Engineering

Course Code: PCC-IT301

Date: 18.08.2025

Time: 15:00 to 15:45

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.

Q.N.		Marks	CO	BT
Q.1	A startup company is developing a mobile-based food delivery application that must be released in six months. The application should allow customers to browse menus, place orders, make online payments, and track deliveries in real time. The development team consists of 8 members with varying experience in mobile app development. As a software engineering consultant, you are tasked to implement SCRUM for this project. Explain how you will apply SCRUM to manage this project by detailing: 1. Stakeholders and User Stories (Generate min 5) 2. Generate Product Backlog 3. Sprint planning and execution 4. Daily Scrum activities	10	CO2	BT3
Q.2	Define Software and Software Engineering. Specify Myths of Practitioner, Customer and Management	5	CO1	BT1
Q.3.	Explain and differentiate Waterfall Model, Iterative model and Prototyping model with example	5	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Basic knowledge and understanding of the analysis and design of complex systems.
CO 2	Ability to apply software engineering principles and techniques.
CO 3	Ability to develop, maintain and evaluate large-scale software systems.
CO 4	To produce efficient, reliable, robust and cost-effective software solutions.
CO 5	Be able to apply coding best practices and perform unit, integration, system, and acceptance testing using established verification and validation techniques